MATH FOUNDATIONS

Linear Algebra Visual Guide

Vectors, matrices, and eigenvalues — the language machine learning speaks.

ANIMATED WALKTHROUGH

Step through pages to play the animation

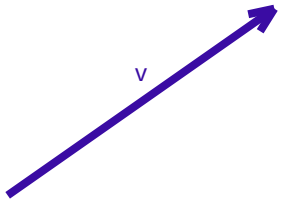
STEP 1/6

Vectors: Direction + Magnitude

An ordered list of numbers is also an arrow in space.

v = [3, 4]

|v| = 5 (3-4-5 triangle)



Add: tip-to-tail. Scale: stretch the arrow. Magnitude: sqrt(sum of squares).

FEATURE VECTORS

[age, income, clicks] — one row of data = one vector

Every ML dataset is a cloud of vectors. The geometry of that cloud is what models learn.

STEP 2/6

Matrices Are Transformations

Multiplying by a matrix moves every vector in space — rotate, scale, shear, project.

Rotation
spins axes

Scaling
stretches axes

Projection
flattens to a line

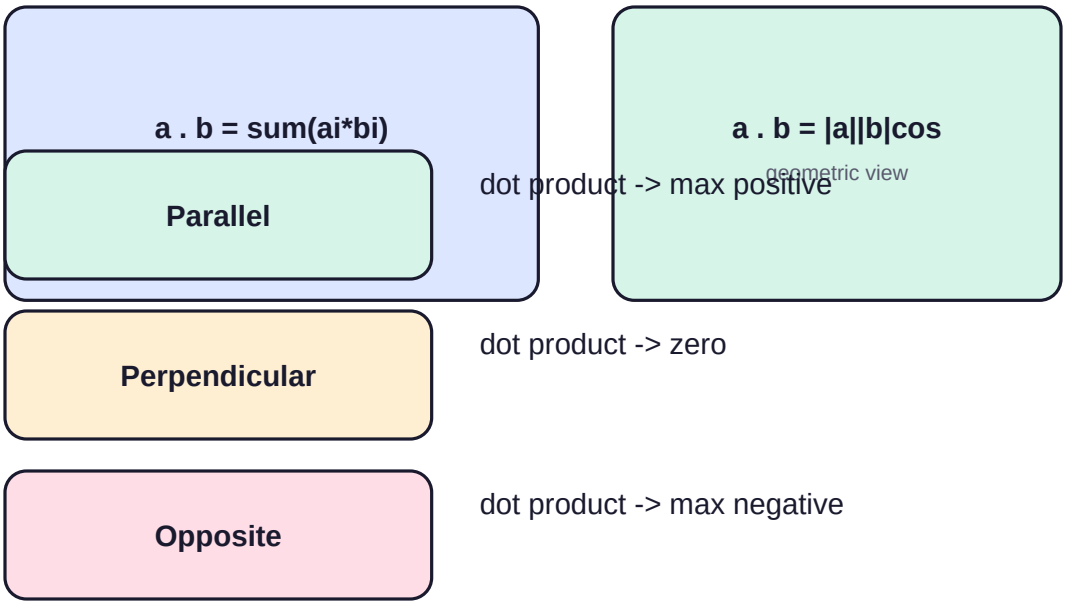
LINEAR LAYERS
a neural net layer IS a matrix multiply plus bias: y = Wx + b

Think function, not table: a matrix maps input space to output space.

STEP 3/6

The Dot Product: Similarity in One Number

a . b = |a||b| cos(theta) — big when vectors point the same way.

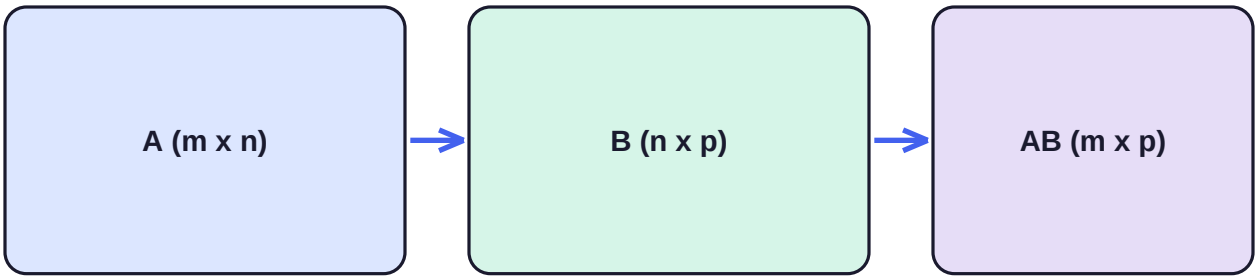


Attention scores, cosine similarity, search ranking: dot products measuring alignment.

STEP 4/6

Matrix Multiplication = Compose Transforms

AB means 'do B first, then A'. Order matters.



Inner dimensions must match: (m x n)(n x p)

NOT COMMUTATIVE

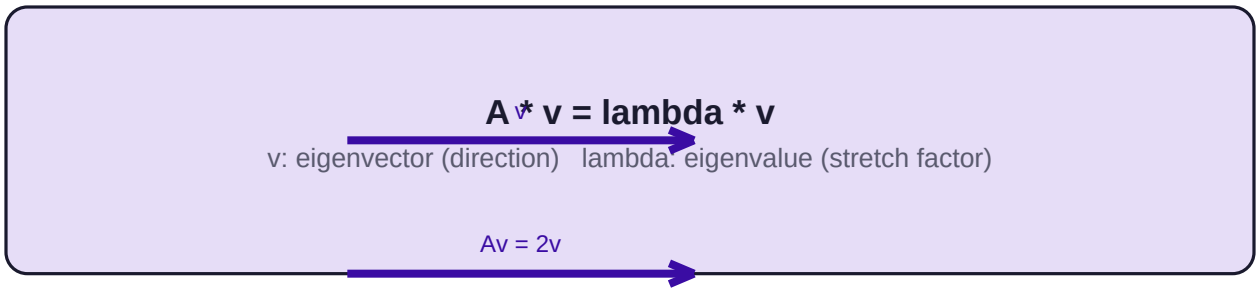
AB != BA in general — like 'rotate then scale' vs 'scale then rotate'

Dimension-check every layer: the inner numbers matching is how you debug shape errors in neural nets.

STEP 5/6

Eigenvectors: The Unmoved Directions

Av = lambda*v — the matrix only stretches these vectors, never rotates them.



Stretch factor 2: same arrow, twice as long. That direction is an eigenvector.

PCA

principal components = eigenvectors of the covariance matrix — the data's true axes

PageRank, PCA, spectral clustering, stability analysis: all eigenvalue problems in disguise.

STEP 6/6

How Linear Algebra Powers ML

The whole field is geometry at scale.

Embeddings

words/images as vectors; meaning = position in space

Cosine similarity

normalized dot product for search & retrieval

Linear regression

projection of y onto the column space of X

SVD / low-rank

compress matrices by keeping top singular directions

Backprop

gradients flow through matrix multiplies via the chain rule

STEP ✓I✓

Key Takeaways

Linear Algebra — Remember

- A vector is a direction plus magnitude; a dataset is a cloud of vectors.
- Matrices are transformations: rotate, scale, shear, project — y = Wx + b is a neural layer.
- The dot product measures alignment: the engine of attention and similarity search.
- Matrix multiplication composes transforms; inner dimensions must match, and order matters.
- Eigenvectors are the directions a matrix only stretches — the basis of PCA and PageRank.
- ML is applied linear algebra: embeddings, projections, and gradients in high-dimensional space.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev